

Head-to-head Evaluation of Translation-First vs Agentic Generate→Validate→Repair Pipelines for Intent-Driven NetOps Changes — Validator-Acceptance Parity under Shared-Candidate Semantics

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Abstract—We report a fully preregistered, artifact-archived paired experiment (CAND-2026-001-prereg-v1) that compared two operational architectures for intent→configuration NetOps changes across heterogeneous vendor CLIs and Mininet-derived topologies: (A) translation-first (constrained vendor DSL → formal verifier → solver) and (B) agentic generate→validate→repair (hosted LLM + deterministic validators). The run declared gpt-5-mini for generation and deterministic_netops_validator_v5 for deterministic end-to-end correctness checks; preregistration used shared_initial_candidate semantics so each task pair received a single LLM candidate shared between arms. Execution completed 240 episodes (120 paired tasks) with model_calls_used = 120 (one shared generation per pair). Deterministic scoring produced contingency counts n11 = 120, n10 = 0, n01 = 0, n00 = 0 (baseline = 120/120; guarded = 120/120). Paired difference (A - B) = 0.00; paired-bootstrap 95% CI = [0.00, 0.00] (10,000 resamples, seed 1729); exact McNemar two-sided p = 1.0. Because generation was intentionally shared, these outcomes measure validator-acceptance parity for identical candidates rather than independent generator or repair robustness. We provide artifact-grounded evidence (execution and analysis manifests, per-task validator traces), an explicit evidence-to-claim mapping table, a nearest-work comparison matrix, and preregistered protocol modifications required to obtain a discriminating head-to-head comparison in follow-ups.

I. INTRODUCTION

Automating intent→configuration translation and safe sequencing of multi-vendor NetOps changes is operationally critical: production networks require both correctness guarantees (reachability, isolation, absence of transient safety violations) and flexibility to express diverse intents across vendor CLIs. Two architectural approaches are common: translation-first pipelines that restrict generation to a vendor DSL and apply formal verification/solvers, and agentic generate→validate→repair pipelines that centralize generation in an LLM and rely on deterministic validators plus bounded repair. Prior literature emphasizes both deterministic guarantees (formal verification) and the promise of LLM-driven flexibility, but direct, preregistered comparisons that produce artifactized per-task validator traces are rare.

This study’s preregistered head-to-head question was: for intent-driven network changes across heterogeneous vendor

CLIs and realistic emulator topologies, which architecture yields higher end-to-end operational correctness (measured by deterministic validation: reachability and policy isolation) — (A) translation-first or (B) agentic generate→validate→repair? The preregistration (CAND-2026-001-prereg-v1) fixed a paired design (N = 120 paired tasks), the primary estimand (paired difference in binary end-to-end correctness A - B), the generator (gpt-5-mini), the deterministic validator (deterministic_netops_validator_v5), and the analysis executor (paired_binary_analysis_v1).

A decisive preregistered design choice was shared_initial_candidate semantics: each paired task used a single LLM candidate generated once and supplied identically to both architectures. This choice deliberately removes generator variance from the confirmatory estimand, turning the experiment into a validator-acceptance parity test for identical candidates rather than an independent generator or repair efficacy comparison. The remainder of the paper (Methods, Results, Discussion) makes that restriction explicit and maps preregistered hypotheses to the measured estimand with artifact references.

II. RELATED WORK

Two research traditions frame our contribution and inform the experiment design. First, formal translation→verification systems focus on deterministic correctness and rely on programmatic abstractions and solver support to produce or certify device configurations. Foundational work on programmatic network verification—SymNet and related abstractions—models network device behavior and forwarding semantics to enable automated reachability analysis and to scale verification via program transformations and symbolic evaluation [10,11]. Complementary work such as "Don’t Mind the Gap" highlights the semantic and implementation gaps that can arise between high-level specification and low-level device behavior and motivates careful abstraction and incremental verification to preserve correctness across translations [10]. More recent formal languages and algebraic frameworks (e.g., Weighted NetKAT and related quantitative verification

approaches) supply compositional semantics and solver interfaces that enable constraint-guided synthesis and metric-aware verification; these tools underpin translation-first pipelines that convert constrained DSL outputs into verifier-checked device artifacts and, when combined with solvers, can produce provably safe configurations under the modeled semantics [5]. Collectively, this literature shows that translation-first architectures trade broader expressivity for stronger, analyzer-level guarantees: by restricting output space and using formal semantics, they can reduce downstream ambiguity and make acceptance decisions deterministic and auditable.

Second, LLM-driven and agentic AIOps work emphasizes flexible, human-friendly intent expression and iterative correction. Recent intent-translation and agentic NetOps studies (for example, INTA at ICNP 2025) investigate conversational intent parsing, in-context translation strategies, and the role of sampling/selection when LLMs produce multiple candidate configurations; these studies document generator variance and the need for downstream validation/repair instrumentation to bound unsafe proposals [3]. Parallel work on verifier-assisted agentic pipelines (e.g., Mahmood & Song, IFIP Networking 2026) demonstrates multi-agent orchestration patterns in which LLM components produce candidates that are iteratively checked by deterministic validators and revised until acceptance; such self-correcting pipelines make the validator the operational authority, instrumenting repair loops and logging verifier traces for auditability [4]. Empirical work on LLM hallucination reduction and structured grounding further argues for hybrid generate→validate→repair designs where validators and constrained prompts limit output-space errors while repair loops address recoverable discrepancies [3,4].

Comparative axes and residual gaps. Prior works differ along a compact set of experimental axes relevant to operational trust: (1) independent generation sampling (whether experiments exercise LLM sampling variance by producing independent candidates per arm), (2) repair-loop instrumentation (whether repair attempts are instrumented and counted), (3) validator acceptance focus (whether evaluation privileges deterministic, formal acceptance criteria versus softer human-centric measures), and (4) emulator/hardware realism (whether experiments validate behavior on real devices or emulators). Formal translation→verification systems excel on axis (3) and (4) when combined with high-fidelity device models, but they commonly restrict generation expressivity and therefore do not explore generator variance on axis (1). Agentic LLM pipelines naturally explore axis (1) and (2) but depend heavily on validator design to bound unsafe outputs. Importantly, few prior studies provide preregistered, paired experimental designs with archived per-task deterministic validator traces that permit reproducible, auditable tests of downstream acceptance parity given identical candidates.

Positioning of the present run. The experiment reported here intentionally targets the intersection of these traditions: it uses LLM generation but freezes generator variance via `shared_initial_candidate` semantics so that the confirmatory estimand isolates downstream validator-acceptance behavior.

This choice draws on the formal literature’s emphasis on deterministic acceptance criteria (we operationalize them via `deterministic_netops_validator_v5`) while adopting artifactized provenance and per-task traces common to reproducible agentic AIOps evaluations. By archiving per-task validator traces, contingency tables, and reconciliation artifacts, the run complements prior translation-first and agentic studies: it does not resolve generator robustness or repair coverage questions under independent sampling (those remain unobserved here), but it supplies a reproducible validator-parity measurement and a concrete protocol that can be modified (e.g., remove `shared_candidate`, enable repair instrumentation, increase `model_calls_used`) to exercise the axes left untested by prior work.

III. METHODOLOGY

This section expands the preregistered and executed methodology with concrete, artifact-grounded implementation and validation details. All referenced artifacts and parameter values are drawn from the archived execution and analysis bundle (paths cited where informative).

Preregistration and frozen contract. The study design was frozen in preregistration CAND-2026-001-prereg-v1. Key preregistered parameters used without post-hoc change: `planned_episode_count` = 240 (120 paired tasks), `model_scope` = {"gpt-5-mini"}, `execution_adapter` = `hosted_netops_gvr_v1`, `generation_semantics` = `shared_initial_candidate`, `maximum_model_calls` = 240, and `validator` = `deterministic_netops_validator_v5` (`validator_version` = 5.0). The preregistered analysis executor was `paired_binary_analysis_v1` with `bootstrap_resamples` = 10000 and `bootstrap_seed` = 1729. The preregistration SHA recorded in the execution manifest is provided for audit.

Task generation and difficulty metadata. Confirmatory tasks were produced deterministically by `deterministic_netops_task_generator_v5`. Each manifest entry (`execution/task_manifest.jsonl`) contains (a) the human-readable intent, (b) canonical reference answer (expected sequence of configuration commands), (c) initial and expected final device states, (d) a `required_sequence` listing required field-level operations, and (e) difficulty metadata fields used for stratification: `changed_interface_count`, `dependency_rule_count`, `level` ∈ {medium, hard, very_hard}, `required_command_count`, and `transient_rule_count`. Representative manifest entries (e.g., `task-000001`) are archived; these entries are the ground truth used by the deterministic validator’s normalization and required-command checks.

Generation orchestration and `shared_initial_candidate` implementation. The orchestration implemented `shared_initial_candidate` as follows: for each paired task the adapter invoked a single LLM generation stage (stage key "`shared_initial_generation`") and persisted the textual candidate. The single stored candidate was then supplied verbatim (no per-arm regeneration) to both the translation-first and agentic downstream arms for

syntactic/semantic processing and deterministic validation. Execution accounting files record `model_calls_used = 120` (one LLM call per task pair) and provider audit metadata (`analysis/deterministic_reconciliation.json`) shows `hosted_model_call_event_count = 120`, `cache_miss_count = 120`, and `cross_condition_cache_key_reuse_observed = false`, indicating each pair used an isolated model invocation without cross-condition reuse.

Model configuration and call semantics. The declared generation model is `gpt-5-mini` as recorded in `execution/model_configuration.json`. Execution parameters recorded in that file include `max_output_tokens = 1500` and `maximum_attempts_per_call = 1`. Provider field `'openai_responses'` documents the execution provenance; no retrieval augmentation was used (`retrieval_augmented_generation = false` in the preregistration). The single-call, single-candidate policy is consistent with the preregistered budget (`model_calls_used <= maximum_model_calls`) and with the planned `shared_candidate` estimand.

Deterministic validator: normalization, constraints, and scoring rules. The `deterministic_netops_validator_v5` implements the preregistered `full_intent_constraint_validation` scorer. Validation is applied in two phases recorded in scoring JSONs: `validation_before` (candidate normalization and difficulty/context diagnostics) and `validation_after` (post-orchestration final state simulation and constraint checks). Concrete checks include: - Syntax and canonical normalization: compute `normalized_response` and canonical ordering when semantically reorderable commands exist; `normalized_response` is archived in `execution/scoring/*.json`. - Required command inclusion: verify that every entry in the task payload's `required_sequence` is present in the candidate (field-level coverage) yielding `required_command_count` and `parsed_command_count`. - Ordering and dependency checks: enforce patterns such as `make-before-break` (uplink switchover), `shutdown_modify_restore` (MTU/VLAN changes requiring admin down), and group constraints (`minimum_active_in_groups`, `final_group_constraints`). Each pattern maps to validator rules that set `violation_codes` and `violation_count`. - Final state simulation: apply the candidate commands to a deterministic emulator model of device states and check that the resultant `final_state` matches `expected_final_state` for the fields constrained by the task, producing `validation_after.valid` boolean recorded per episode.

Scoring semantics and success definition. Per preregistration, a task episode is scored success (binary 1) iff `validation_after.valid == true` and `scoring_rule == "full_intent_constraint_validation"`. The scoring JSONs include `score_reason_code` (e.g., `VALID_CONFIGURATION`) and `repair_applied` flag. All per-episode scoring artifacts (`execution/scoring/task-000***-*.json`) are the canonical record of the validator decision and include `raw_response_sha256`, `shared_initial_candidate_sha256`, and `validator_trace` with parsed counts and normalized configurations.

Repair policy and instrumentation. The preregistered `transformation_scope` allowed a single repair iteration per task (`maximum_repair_calls_per_task = 1`). The orchestration records `repair_provider_trace_path` and `repair_provider_trace_sha256` when repair is requested and applied. In this execution, aggregated `repair_applied` count = 0 across `execution/scoring/*.json` (`repair_applied == false` for every episode), therefore the repair loop remained uninvoked; this observation is archived and referenced in analysis artifacts.

Contamination checks, missingness, and treatment rules. The preregistered `contamination_plan` included (i) a simulator fidelity suite that compares emulator outputs to held-out public device examples and flags tasks with >5% disagreement, (ii) grammar coverage/unit tests for vendor CLI constructs, and (iii) execution isolation (explicit model context resets). `Analysis/contamination_summary.csv` is empty for this run (no flags). Missingness rules allowed two retries on transient provider failures and specified exclusion only when both pipeline runs were lost for a pair; the execution manifest reports `failed_episode_count = 0` and `missingness_summary` indicates `complete_pairs = 120`.

Analysis executor and inferential procedures. Analysis followed `paired_binary_analysis_v1` exactly as preregistered. The executor computed the 2×2 paired contingency table ($n_{11}, n_{10}, n_{01}, n_{00}$), the paired difference in success rates ($A - B$), bootstrap percentile 95% CI using `bootstrap_resamples = 10000` and `bootstrap_seed = 1729` (both recorded in `analysis/analysis_log.jsonl`), and an exact McNemar two-sided test for the paired binary outcome. Analysis artifacts (`analysis/paired_contingency_table.csv`, `analysis/condition_summary.csv`, `analysis/analysis_log.jsonl`) and `deterministic_reconciliation.json` provide reproducible inputs and outputs for independent recomputation.

Reproducibility and artifact references. Every described artifact is archived in the execution bundle and referenced in the execution manifest and analysis artifacts. Important file paths used by reviewers/readers: `execution/execution_manifest.json` (execution accounting and preregistration `sha256`), `execution/model_configuration.json` (`gpt-5-mini` declaration), `execution/task_manifest.jsonl` (task payloads and difficulty fields), `execution/responses/*` (shared and per-condition textual candidates), `execution/scoring/*.json` (per-episode validator traces and scoring decisions), `analysis/paired_contingency_table.csv` and `analysis/condition_summary.csv` (primary outcomes), and `analysis/analysis_log.jsonl` (bootstrap parameters). These artifacts collectively document the operational semantics: single generation per pair $\rightarrow$ identical candidate delivered to both arms $\rightarrow$ deterministic normalization and constraint validation $\rightarrow$ binary success decision archived per episode.

Protocol limitations relevant to methodology. The `shared_initial_candidate` semantics were a preregistered design choice and implemented deterministically; it removes generator sampling variance and the operational effect of independent repairs. The methodology above therefore describes an exactly specified validator-acceptance parity

measurement: it assesses downstream orchestration and validator agreement on identical textual candidates rather than independent generator robustness. Practical follow-ups to test independent generation and repair require concrete protocol changes (remove shared_candidate, increase model_calls_used to generate independent candidates per arm, enable repair calls) that are specified in the manuscript’s Discussion and documented in the preregistration as amendable, auditable modifications.

IV. RESULTS

Execution accounting and primary numeric outcomes. The execution manifest reports planned_episode_count = 240, completed_episode_count = 240, failed_episode_count = 0 and model_calls_used = 120 (one shared generation per paired task). Provider audit and deterministic reconciliation confirm hosted_model_call_event_count = 120, cache_miss_count = 120, cross_condition_cache_key_reuse_observed = false, and complete_pair_count = 120, consistent with the preregistered shared_initial_candidate execution semantics.

Deterministic scoring and contingency counts. The complete paired contingency is n11 = 120, n10 = 0, n01 = 0, n00 = 0. Marginal success rates are baseline = 120/120 = 1.0 and guarded = 120/120 = 1.0 (analysis/condition_summary.csv records these counts explicitly). The paired difference (A - B) equals 0.00. The paired bootstrap percentile 95% CI computed with 10,000 resamples and bootstrap_seed = 1729 is [0.00, 0.00]; the exact McNemar two-sided p-value is 1.0. These numbers are direct outputs of the preregistered paired_binary_analysis_v1 executor and are archived in analysis/paired_contingency_table.csv and analysis/analysis_log.jsonl.

Repair instrumentation and secondary estimands. Across all 240 episodes the archived per-task scoring JSONs show repair_applied = false (aggregated repair_applied count = 0). Consequently, preregistered secondary estimands concerning repair coverage and mean repair iterations are unobserved in this execution. The absence of repair events is reflected uniformly in execution/scoring/*.json where validator_trace.repair_applied = false and repair_provider_trace_path is null.

Representative per-task diagnostics. Representative artifacts illustrate the validator acceptance logic and task difficulty span. Example highlights from archived scoring JSONs:

- pair-000001 (task-000001): difficulty.level = "medium"; parsed_command_count = 4, required_command_count = 4. The recorded normalized_reference_answer differs in command ordering from the normalized_response (vlan and mtu lines swapped), yet validation_after.valid = true — demonstrating that the validator’s normalization and dependency checks accept ordering variants when required commands and final state constraints are satisfied.
- pair-000002 (task-000002): difficulty.level = "hard"; parsed_command_count = 2, required_command_count = 2; normalized_response matches the reference exactly and validation_after.valid = true.
- pair-000003 (task-000003):

difficulty.level = "hard"; parsed_command_count = 6, required_command_count = 6; again validation_after.valid = true. These representative traces show tasks across medium→hard levels, with parsed_command_count and required_command_count recorded per task and used by the deterministic validator to assert correctness.

Contamination, missingness, and execution integrity diagnostics. The preregistered contamination checks produced no flags: analysis/contamination_summary.csv is empty. Missingness diagnostics record complete_pairs = 120 and zero failed or unscorable episodes (analysis/missingness_summary.csv). The deterministic_reconciliation artifact confirms episode and pair accounting consistency and that all deterministic consistency checks passed.

Interpretation of the degenerate concordance. The observed perfect concordance (all pairs accepted by both downstream arms) yields a degenerate statistical picture: the paired difference estimate is point-mass zero, the bootstrap CI collapses to [0.00,0.00], and McNemar’s test yields p = 1.0. These statistics correctly quantify uncertainty given the observed data but arise from the preregistered decision to share LLM candidates between arms. Because generator variance was removed by design, the confirmatory outcome answers the narrower question of validator-acceptance parity for identical candidates rather than the broader question of independent generator or repair superiority. This is consistent with the preregistration which specified shared_initial_candidate semantics and is explicitly recorded in execution/model_configuration.json and the execution manifest.

Why repairs were not applied in this run (artifact-grounded observation). The archived per-task traces indicate that shared LLM candidates either matched the canonical normalized reference (exact match) or satisfied the validator’s normalization and dependency checks (ordering-insensitive acceptance). Representative normalized_response and normalized_reference_answer fields in execution/scoring/task-000XXX-*.json demonstrate this pattern: validator normalization reconciled minor syntactic/order variations and all required commands and final state constraints were present, so no repair loop was triggered. This operational detail explains the observed repair_applied = false counters without invoking any unarchived assumptions.

Consequences for preregistered hypotheses and next steps. H1 (paired difference in end-to-end correctness A - B) is measured here but under the restricted estimand implied by shared_initial_candidate semantics (validator-acceptance parity). H2 (agentic repair coverage advantage) is unobserved because no repair events occurred. The artifact bundle and execution accounting provide explicit levers for a discriminating follow-up (remove shared candidate to reintroduce generator variance, increase model_calls_used to produce independent per-arm generations, enable and instrument repair attempts) — each such change would produce verifiable differences in archived metrics (hosted_model_call_event_count, nonzero repair_applied flags, increased model_calls_used)

that would support testing the originally preregistered repair-and-generation hypotheses.

V. DISCUSSION

What the run measures — and what it does not. Because `shared_initial_candidate` semantics were used, the preregistered confirmatory estimand reduces to validator-acceptance parity: given identical LLM candidates, do downstream orchestration and verification paths differ in deterministic validator acceptance? The observed complete concordance ($n_{11} = 120$) answers this restricted question: both validators accepted the identical candidates for every paired task. The run does not exercise independent generator variance, sampling effects, or repair efficacy; therefore claims about which architecture would produce more correct initial candidates or achieve higher repair coverage when generating independently are untested.

Evidence-to-claim mapping (compact table). Table 2 (in manuscript) maps preregistered hypotheses/estimands to measured observables and archived artifacts. Example mappings: H1 (paired difference $A - B$) $\rightarrow$ measured estimand = validator-acceptance parity under `shared_initial_candidate`; supporting artifacts: `analysis/paired_contingency_table.csv` and `analysis/condition_summary.csv` ($n_{11}=120, n_{10}=0, n_{01}=0, n_{00}=0$); verdict: tested but restricted. H2 (agentic repair coverage advantage) $\rightarrow$ observed `repair_applied` count = 0 in `execution/scoring/*.json`; verdict: unobserved in this execution.

Operational implications and protocol guidance for discriminating comparisons. The artifact bundle documents a reproducible protocol for validator regression and orchestrator equivalence checks under fixed candidates. To test independent-generation and repair differences the preregistered protocol must be modified (and these modifications can be preregistered and audited): (i) remove `shared_initial_candidate` so each arm receives an independent LLM generation per pair (`model_calls_used` would rise from 120 to 240 for initial candidates); (ii) allow multiple sampled initial generations per arm to estimate generator variance; (iii) enable and instrument repair iterations so `repair_applied` counters become informative; (iv) increase emulator fidelity or include hardware runs to reduce simulator-to-hardware contamination risk. Each modification yields verifiable changes in execution artifacts (`model_calls_used`, `hosted_model_call_event_count`, nonzero `repair_applied` flags) and therefore supports reproducible discrimination across architectures.

Threats to validity (concise). Internal validity: `shared_candidate` removes generator variance by design; this is a feature not a flaw but narrows the estimand. Construct validity: deterministic validator acceptance operationalizes a conservative correctness definition but omits human-facing properties (explainability, maintainability). External validity: synthetic tasks and Mininet-derived emulation plus a single model (gpt-5-mini) and single validator (`deterministic_netops_validator_v5`) limit generality.

Conclusion validity: degenerate concordance (all pairs concordant) produces point mass CI and $p = 1.0$; these statistics correctly describe the observed data but do not imply production-level equivalence under independent operation.

Reproducibility and audit artifacts. All primary artifacts are archived and referenced by `execution/execution_manifest.json`: preregistration (`CAND-2026-001-prereg-v1`), `execution/model_configuration.json` (`declared_model_version = gpt-5-mini`), `execution/task_manifest.jsonl`, `execution/responses/*`, `execution/scoring/*.json`, `analysis/paired_contingency_table.csv`, `analysis/condition_summary.csv`, `analysis/analysis_log.jsonl`, and `analysis/deterministic_reconciliation.json`. Per-task validator traces (`execution/scoring/task-000XXX-*.json`) are the canonical records for each scored decision. The analysis executor parameters (`bootstrap_resamples = 10000`, `bootstrap_seed = 1729`) are recorded in `analysis/analysis_log.jsonl` and enable deterministic recomputation of the reported CI.

VI. LIMITATIONS

- Preregistered `shared_initial_candidate` semantics were used; this intentionally removes generator variance and prevents this run from assessing independent generation robustness differences between architectures.
- The task corpus was synthetic and executed in an emulator (Mininet-derived topologies and public vendor example grammars); emulator-to-hardware divergence limits external validity to production devices.
- A single LLM (gpt-5-mini) and a single deterministic validator (`deterministic_netops_validator_v5`) were used; results do not imply multi-model or multi-validator generality.
- No repairs were applied in this run (`repair_applied = false` in per-task traces); preregistered secondary estimands about repair coverage remain unobserved for this execution.
- The binary deterministic validator acceptance metric does not capture operator-facing properties such as explanation quality, maintainability, or human trust, which matter in production but were outside the metric used here.

VII. CONCLUSION

We executed a preregistered, artifactized paired experiment comparing translation-first and agentic generate $\rightarrow$ validate $\rightarrow$ repair NetOps pipelines under `shared_initial_candidate` semantics. For 120 paired tasks deterministic validators accepted identical LLM candidates in both arms for every pair ($n_{11} = 120, n_{10} = n_{01} = n_{00} = 0$), yielding paired difference 0.00 (bootstrap 95% CI [0.00, 0.00], McNemar $p = 1.0$). No repairs were applied (`repair_applied = false` for all episodes); preregistered hypotheses about repair coverage are therefore unobserved. The run precisely measures validator-acceptance parity under controlled shared candidates and provides a reproducible artifact bundle and

explicit protocol modifications for future discriminating comparisons that exercise independent generation and repair coverage.

DISCLOSURE STATEMENT

This research was executed autonomously after the autonomy lock under the frozen master prompt archived with the run provenance. Preregistration: CAND-2026-001-prereg-v1 (archived and used to freeze the design; preregistration_sha256 recorded in the execution manifest). Generation used gpt-5-mini (declared_model_version in execution/model_configuration.json); provider record 'openai_responses'. The hosted adapter family was hosted_netops_gvr_v1. Deterministic artifacts included deterministic_netops_task_generator_v5 and deterministic_netops_validator_v5 (validator_version 5.0). The run deliberately used the preregistered shared_initial_candidate generation semantics: both pipeline arms received the same initial LLM candidate per task; execution accounting shows model_calls_used = 120 (one shared generation per pair) while planned episodes were 240. Primary analysis used paired bootstrap percentile CIs (10,000 resamples, seed 1729) and an exact McNemar two-sided test executed by paired_binary_analysis_v1. No human manually edited, rewrote, or post-processed technical content after the autonomy lock. The Research Director selected the broad topic family and constrained the initial master prompt prior to the autonomy lock; the autonomous run performed literature review, final research-question selection, preregistration, code generation, experiment execution, analysis, peer review, and manuscript drafting after the lock. Immutable reference to the initial master prompt: provenance/master_prompt.txt (SHA-256 = 1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdf1582bb39b15ba). The full execution and analysis artifact bundle is archived and referenced by the execution manifest included with this submission.

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